Bahaa Mohammad

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U.S. Citizen | Languages: English, Arabic

Meet me :)

Education

University of Iowa

Master of Science in Electrical and Computer Engineering, Accelerated BS/MS — **Expected May 2027** • GPA 3.34
Bachelor of Science in Engineering in Biomedical Engineering — **May 2026** • GPA 3.15

LinkedIn Learning: Project Management Foundations; Negotiation Skills; Agile Development Practices.

Work Experience

Software Engineer — June 2026 – Present
Voss Lab (HBC)

- Built a **Python**/**R** pipeline (**JATOS** → auto **GGIR** → **REDCap** QC → debrief reports) for **295** subjects, replacing spreadsheet triage in cohort review.
- Automated **JATOS** pulls, **Globus** transfers, and **HPC** job submission so weekly pipelines sync and process **110 GB+** of multi-source study data end-to-end.

Systems Engineer — May 2025 – Present
IHuman Technology (Santos)

- Designed an **Oracle S3**-backed **C#** release/updater for **5** simulation apps with **GitHub Actions** CI/CD (merge → runner rebuild → version/size JSON → S3 publish); shared common assets cut combined install from ∼**7.5 GB** to a **2.2 GB** cap (∼**70%**).
- Packaged delivery with **Docker**; migrated UI rendering to **C++** Diligent Engine and wired **C#/.NET** native interop so release builds exercise the full render path; cleared cross-platform defects and confirmed regression coverage before shipping each **GitHub Actions** release to **Oracle S3**.

Research Software Engineer — March 2026 – May 2026
Iowa Technology Institute (I-STRIKE Santos)

- Embedded a **MuJoCo** physics update loop inside Santos (**215** DOFs) via legacy **C#/C++** interop; fixed zero-mass bodies, frame mismatches, and anchors so contact simulation loads and runs.

Software Engineer — April 2024 – March 2026
Iowa Technology Institute (VSR / Santos DHM)

- Owned day-to-day development on a **140+**-project **C++/C#/.NET/Fortran** Santos suite used daily by the research software team, shipping multi-language features into production research builds.
- Built and maintained **GitLab CI/CD** across the multi-language monorepo so merge builds stayed green, and kept Santos kinematics-to-graphics pipelines aligned so **C++/C#/.NET** visualization matched live simulation state.

Software Projects

Lumora — **Go**, **React**, **TypeScript**

- Built a multi-tenant Go/React platform powering production clients Alamir Auto and Franks Hot Chicken—shared auth, payments, media, and tenant-isolated data. lumorasites.com, alamirauto.com, frankshotchickensmashburger.com

BME Senior Design — Hip Contact Stress Mapping — **MATLAB**, **C++**, **.NET MAUI**

- Tech lead (∼**98%** implementation); owned FRs/implementation narrative; solo Orthopedics demo for ∼**18** clinicians; technical narrative at Prototype Day / senior-design night.

SlamNav: GPS-Denied Indoor Mapping — **KMP**, **C++**, **Swift**, **Objective-C**, **Java**

- Architecture lead for a **3**-person team; shipped Android+iOS on time with full marks after keeping dual-platform when Android-only would have stranded teammates; demoed at ECE Modern Marvels.

Hardware Projects

Remote ECG Monitoring System (BME Systems) — **C++; Arduino, ESP32 Wi-Fi**

- Built a 1-lead analog front-end (instrumentation amp, RLD, bandpass, DC level-shift) and C++ R-peak/BPM pipeline; pivoted ESP32 Wi-Fi to wired Arduino after RF EMI corrupted the adaptive-threshold detector.

CircSim: Embedded Circuit Solver — **C; TI-84 virtual + physical**

- Memory-conscious nodal-analysis solver in C under TI-84 limits; debugged on TI virtual calculator and ran on a physical TI-84.

Leadership

Kitchen Leader — April 2023 – July 2024; July 2025 – September 2026
Chipotle Mexican Grill

- Led peak-rush kitchen ops: assigned prep/line roles, paced throughput, and held food safety and cleanliness under pressure.
- Owned morning commercial review (critical-item variance under **0.2%**) and coached guest recovery under pressure.

Publications, Participation

- Abdel-Malek et al. & **Mohammad, B.** (2025). Santos DHM / Army ACFT. 9th Int. DHM Symp.
- Ali, M. S., **Mohammad, B.**, et al. (2025). Self-supervised spiral VAE, upper-airway MRI. ISMRM Workshop.
- **MSA** (2 year) Vice President • **NSBE** (3 year) Senator